

Boyce-Codd Normal Form (BCNF) -- Relational Database Design

Definition: A relation is in Boyce-Codd Normal Form (BCNF) (also known as 3.5NF) if and only if for every non-trivial functional dependency $X \rightarrow Y$, X is a strict Super Key.

1. Detailed Technical Explanation

BCNF is a stricter version of 3NF that handles cases where a relation has multiple overlapping candidate keys.

3NF vs BCNF Comparison:

Property	3NF	BCNF
Allowed Condition $X \rightarrow Y$ **	X is Super Key **OR Y is Prime Attri	X MUST be a Super Key (No exceptions!)
Dependency Preservation	Always Guaranteed	Not always guaranteed
Redundancy	Allows minor non-key redundancy	Completely eliminates functional redun

Conversion Example to BCNF

Relation ADVISOR(Student_ID, Subject, Advisor_Name)

Relation Status:

BCNF Decomposition:

2. Core Concepts & Memory Keywords

- * Strict Super Key Rule: Every determinant in a functional dependency must be a super key.

3. Must-Write Points for Exams

- * BCNF eliminates all redundancy resulting from functional dependencies.

4. Quick Recall Flow

3NF Relation $\rightarrow$ Inspect all FDs ($X \rightarrow Y$) $\rightarrow$ Verify if X is Super Key $\rightarrow$ Decompose non-superkey determinants $\rightarrow$ BCNF Achieved